Inverse & Composite Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Composing functions**

Given $f(x) = \sqrt{x+4}$ and $g(x) = x^2 - 4$, find $(f \circ g)(2)$ and state the domain restriction that makes $f \circ g$ defined.

$(f \circ g)(2) = f(g(2)) = f(0) = \sqrt{0+4} = 2$. In general, $f \circ g$ requires $g(x) + 4 \geq 0$, i.e. $x^2 \geq 0$, which holds for all real x — so $f \circ g$ is defined everywhere.

2 **[Foundational]** • **Target: Decomposing a composite function**

Write $h(x) = (3x - 5)^4$ as a composition $h(x) = f(g(x))$ for two simpler functions f and g.

Let $g(x) = 3x - 5$ (inner function) and $f(x) = x^4$ (outer function). Then $f(g(x)) = (3x - 5)^4 = h(x)$.

3 **[Foundational]** • **Target: Decomposing a composite function**

Write $h(x) = \sqrt{x^2 + 1}$ as a composition of two functions.

Let $g(x) = x^2 + 1$ and $f(x) = \sqrt{x}$. Then $f(g(x)) = \sqrt{x^2 + 1} = h(x)$.

4 **[Foundational]** • **Target: Verifying inverse functions**

Verify algebraically that $f(x) = \frac{x-2}{3}$ and $g(x) = 3x + 2$ are inverses of each other.

$f(g(x)) = \frac{(3x+2)-2}{3} = \frac{3x}{3} = x$. $g(f(x)) = 3\left(\frac{x-2}{3}\right) + 2 = (x - 2) + 2 = x$. Since both compositions return x, f and g are inverses.

5 **[Foundational]** • **Target: Verifying inverse functions**

Find the inverse of $f(x) = 2x^3 + 1$, and verify that $f(f^{-1}(x)) = x$ for one value, say $x = 17$.

Solve $y = 2x^3 + 1$ for x: $x^3 = \frac{y-1}{2}$, so $f^{-1}(x) = \sqrt[3]{\frac{x-1}{2}}$. Check at $x = 17$: $f^{-1}(17) = \sqrt[3]{8} = 2$, and $f(2) = 2(8) + 1 = 17$ ✓.

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6 [Foundational] • Target: Visualizing inverse functions as reflections

The graph shows $f(x) = x^2$ (restricted to $x \geq 0$) and its inverse $f^{-1}(x) = \sqrt{x}$, which are reflections of each other across the line $y = x$.

a State the domain restriction placed on $f(x) = x^2$ needed for it to have an inverse, and explain why.

$f(x) = x^2$ must be restricted to $x \geq 0$ (or $x \leq 0$) because the full parabola fails the horizontal line test — without restricting the domain, f is not one-to-one and has no inverse function.

b Confirm from the graph that f and f^{-1} intersect on the line $y = x$, and find that intersection point algebraically. Setting $f(x) = x$: $x^2 = x$, so $x^2 - x = 0$, $x(x - 1) = 0$, giving $x = 0$ or $x = 1$. Both points $(0, 0)$ and $(1, 1)$ lie on $y = x$, matching where the two curves cross in the graph.

(a) $f(x) = x^2$ must be restricted to $x \geq 0$ (or $x \leq 0$) because the full parabola fails the horizontal line test — without restricting the domain, f is not one-to-one and has no inverse function.; (b) Setting $f(x) = x$: $x^2 = x$, so $x^2 - x = 0$, $x(x - 1) = 0$, giving $x = 0$ or $x = 1$. Both points $(0, 0)$ and $(1, 1)$ lie on $y = x$, matching where the two curves cross in the graph.

7 [Foundational] • Target: Domain and range of composite and inverse functions

Given $f(x) = \frac{1}{x - 3}$ and $g(x) = x + 1$, find the domain of $(f \circ g)(x)$.

$(f \circ g)(x) = f(x + 1) = \frac{1}{(x + 1) - 3} = \frac{1}{x - 2}$. This is undefined at $x = 2$, so the domain is all reals except $x = 2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Domain and range of composite and inverse functions

If f has domain $[1, \infty)$ and range $[0, \infty)$, state the domain and range of f^{-1}.

The domain and range swap for an inverse function: f^{-1} has domain $[0, \infty)$ and range $[1, \infty)$.

9 [Intermediate] • Target: Applying composite and inverse functions in context

A temperature-conversion function $C(F) = \frac{5}{9}(F - 32)$ converts Fahrenheit to Celsius. Find $C^{-1}(F)$ and interpret what it computes.

Solve $C = \frac{5}{9}(F - 32)$ for F: $\frac{9C}{5} = F - 32$, so $F = \frac{9}{5}C + 32$. This is C^{-1}, which converts a Celsius temperature back to Fahrenheit.

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10 [Intermediate] • Target: Using the horizontal line test

Determine whether $f(x) = x^3 - 4x$ has an inverse function over its entire domain. Justify using the horizontal line test.

Since $f(x) = x^3 - 4x$ has multiple turning points (a cubic with two real critical points), a horizontal line such as $y = 0$ intersects the graph at more than one point ($x = 0, \pm 2$), failing the horizontal line test. So f does not have an inverse function over its entire domain.

11 [Intermediate] • Target: Using the horizontal line test

Explain why $f(x) = 5x - 2$ passes the horizontal line test for all real x, and find its inverse.

$f(x) = 5x - 2$ is a line with nonzero slope, so it is strictly increasing and every horizontal line crosses it exactly once — it passes the horizontal line test everywhere. Solving $y = 5x - 2$ for x: $f^{-1}(x) = \frac{x+2}{5}$.

12 [Intermediate] • Target: Composite functions with restricted domains

Given $f(x) = \frac{1}{x}$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and state its domain.

$(f \circ g)(x) = f(x-4) = \frac{1}{x-4}$. This is undefined at $x = 4$, so the domain is all reals except $x = 4$.

13 [Intermediate] • Target: Domain and range of composite and inverse functions

Find the inverse function of $f(x) = \frac{2x-1}{x+3}$.

Step 1: Set $y = \frac{2x-1}{x+3}$ and swap variables x and y: $x = \frac{2y-1}{y+3}$ Step 2: Multiply both sides by $(y+3)$: $x(y+3) = 2y - 1 \Rightarrow xy + 3x = 2y - 1$ Step 3: Group terms containing y on one side: $xy - 2y = -3x - 1 \Rightarrow y(x-2) = -(3x+1)$ Step 4: Solve for y: $y = \frac{-3x-1}{x-2} = \frac{3x+1}{2-x} \quad (x \neq 2)$

14 [Intermediate] • Target: Applying composite and inverse functions in context

Determine whether the function $g(x) = \frac{x^3}{x^2+4}$ is even, odd, or neither.

Evaluate $g(-x)$ by substituting $-x$ for every occurrence of x: $g(-x) = \frac{(-x)^3}{(-x)^2+4} = \frac{-x^3}{x^2+4} = -\left(\frac{x^3}{x^2+4}\right) = -g(x)$ Because $g(-x) = -g(x)$ for all $x \in \mathbb{R}$, $g(x)$ is an odd function (symmetric with respect to 180° rotation about the origin).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Using the horizontal line test • Cross-Lesson: 2.01 Exponential Functions & Growth Models

A function is defined piecewise: $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$. What is the value of $f(-3) + f(2)$?

Evaluate each term according to its interval condition: For $x = -3 < 0$: $f(-3) = (-3)^2 = 9$. For $x = 2 \geq 0$: $f(2) = 2(2) + 1 = 5$. Sum: $f(-3) + f(2) = 9 + 5 = \mathbf{14}$.

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16 [Challenging] • Target: Composite functions with restricted domains • Cross-Lesson: 2.02 Logarithmic Functions & Properties

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

17 [Challenging] • Target: Composing functions • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

[Exponential & Logarithmic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 [Challenging] • Target: Decomposing a composite function • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

[Functions & Their Graphs · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

19 [Challenging] • Target: Verifying inverse functions • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

[Polynomial Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 [Challenging] • Target: Visualizing inverse functions as reflections • Cross-Lesson: 4.03 Parametric Equations & Modeling

[Trigonometric Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Composing functions

Given $f(x) = -7x$, find the slope and y-intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).

a. $(-7, 0)$

b. $\left(-\frac{1}{7}, 0\right)$ ✓

c. $\left(\frac{1}{7}, 0\right)$

d. $\left(\frac{5}{14}, 0\right)$

Solve $y = -7x$ for x: $x = \frac{y}{-7}$. So $f^{-1}(x)$ has slope $\frac{1}{-7} = -\frac{1}{7}$ and intercept 0: $\left(-\frac{1}{7}, 0\right)$.

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22 [Foundational] • Target: Decomposing a composite function

Determine whether $f(x) = x^3 - 6x$ has an inverse function over its entire domain, using the horizontal line test.

a. Has an inverse function over its entire domain (strictly increasing)

b. **Does NOT have an inverse function over its entire domain (not one-to-one)** ✓

c. Has an inverse only on the interval $x \geq 0$

d. Cannot be determined without a graph

Since $b = -6 < 0$, $f(x) = x^3 - 6x$ has two turning points (critical points where $3x^2 - 6 = 0$), so some horizontal lines cross the graph more than once. Does NOT have an inverse function over its entire domain (not one-to-one).

23 [Foundational] • Target: Verifying inverse functions

What is the domain and range of $y = \tan^{-1}(x)$?

a. Domain $(-\infty, \infty)$, range $(-\infty, \infty)$

b. Domain $(-\infty, \infty)$, range $[0, \pi]$

c. Domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, range $(-\infty, \infty)$

d. **Domain $(-\infty, \infty)$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$** ✓

$\tan^{-1}(x)$ has domain $(-\infty, \infty)$ (restricted input) and range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (restricted principal output).

24 [Foundational] • Target: Visualizing inverse functions as reflections

What is the domain and range of $y = \sin^{-1}(x)$? (variant 5)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, range $[-1, 1]$

c. **Domain $[-1, 1]$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$** ✓

d. Domain $(-\infty, \infty)$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\sin^{-1}(x)$ has domain $[-1, 1]$ (restricted input) and range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (restricted principal output).

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25 [Foundational] • Target: Domain and range of composite and inverse functions

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown in the standard (x, y) coordinate plane below. A horizontal line $y = k$ intersects the graph of $f(x)$ at two points that are 8 units apart. What is the value of k?

a. 12 ✓

b. 8

c. 4

d. -3

We are given the quadratic function $f(x) = x^2 - 6x + 5$ and a horizontal line $y = k$ that intersects the parabola at two points separated by a horizontal distance of 8. **Method 1: Using the axis of symmetry** The axis of symmetry for a parabola in standard form $y = ax^2 + bx + c$ is given by: $x = -\frac{b}{2a} = -\frac{-6}{2(1)} = 3$ Because parabolas are symmetric about their axis of symmetry, the two intersection points on the horizontal line $y = k$ must lie equidistant from $x = 3$. Since the total dist.

26 [Foundational] • Target: Applying composite and inverse functions in context

Consider the function $f(x) = 2x + 5$. Which of the following represents the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{x-5}{2}$ ✓

b. $f^{-1}(x) = \frac{x+5}{2}$

c. $f^{-1}(x) = \frac{2}{x} - 5$

d. $f^{-1}(x) = 2x - 5$

To find the inverse function $f^{-1}(x)$, we first replace $f(x)$ with y: $y = 2x + 5$. Next, we swap x and y: $x = 2y + 5$. Now, we solve for y to isolate the inverse function. Subtract 5 from both sides: $x - 5 = 2y$. Divide both sides by 2: $\frac{x-5}{2} = y$. Therefore, the inverse function is $f^{-1}(x) = \frac{x-5}{2}$.

27 [Foundational] • Target: Using the horizontal line test

Find the inverse function: if $y = \sin x$, then $x =$

a. $\sin^{-1} y$ for $y \in [0, 1]$ only

b. $\cos^{-1} y$

c. π/y

d. $\arcsin y$ **for** $y \in [-1, 1]$ ✓

Applying Trigonometric Functions principles, Definition of arcsine. Therefore, the verified correct option is $\arcsin y$ for $y \in [-1, 1]$.

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28 [Foundational] • Target: Composite functions with restricted domains

A ramp rises 3 m over a horizontal distance of 20 m. Find its angle of inclination, to the nearest tenth of a degree.

a. 8.5° ✓

b. 81.5°

c. 13.5°

d. 17.1°

In the right triangle, opposite $= 3$, adjacent $= 20$, so the angle is $\tan^{-1}\left(\frac{3}{20}\right) \approx 8.5^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Composing functions

Find the inverse of $f(x) = \frac{3x+2}{x-1}$ and state its domain.

a. $f^{-1}(x) = \frac{x+2}{x-3}$; **domain: all reals except** $x = 3$ ✓

b. $f^{-1}(x) = \frac{3x-2}{x+1}$

c. $f^{-1}(x) = \frac{x+2}{x+3}$

d. $f^{-1}(x) = \frac{x-2}{x+3}$

Swap: $x = \frac{3y+2}{y-1} \Rightarrow x(y-1) = 3y+2 \Rightarrow xy - x = 3y+2 \Rightarrow y(x-3) = x+2 \Rightarrow y = \frac{x+2}{x-3}$.

30 [Intermediate] • Target: Decomposing a composite function

The graph of a polynomial function $f(x)$ of degree 4 is shown in the figure below. The graph is tangent to the x-axis at $(-2, 0)$, intersects the x-axis at $(1, 0)$ and $(3, 0)$, and crosses the y-axis at $(0, -6)$. A new polynomial $Q(x)$ is defined by $Q(x) = f(x) + \frac{1}{2}x^3$. When $Q(x)$ is divided by $x^2 - 4$, the remainder is the linear polynomial $R(x)$. What is the value of $R(5)$?

a. 24 ✓

b. 18

c. 17

d. 14

We solve the problem in two key phases: identifying the polynomial $f(x)$ from its graphical features, and applying the Polynomial Remainder Theorem to determine $R(x)$. **Step 1: Determine $f(x)$** - Because the graph is tangent to the x-axis at $x = -2$, $(x + 2)$ is a factor with an even multiplicity of at least 2. - Because the graph crosses the x-axis at $x = 1$ and $x = 3$, $(x - 1)$ and $(x - 3)$ are factors with odd multiplicities of at least 1. - Since $f(x)$ is of degree 4, the sum of multipliciti.

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31 [Intermediate] • Target: Verifying inverse functions

The graph of a polynomial function $f(x)$ is shown below. Which of the following could be the equation for $f(x)$?

a. $f(x) = (x + 2)(x - 1)(x - 3)$ ✓

b. $f(x) = (x - 2)(x + 1)(x + 3)$

c. $f(x) = -(x + 2)(x - 1)(x - 3)$

d. $f(x) = (x + 2)(x - 1)^2(x - 3)$

To determine which polynomial function matches the graph, we analyze the intercepts, multiplicities, and end behavior: 1. **Identify the x-intercepts (zeros):** The graph crosses the x-axis at $x = -2$, $x = 1$, and $x = 3$. By the Factor Theorem, if c is a zero of $f(x)$, then $(x - c)$ is a factor. Therefore, the corresponding linear factors are: $(x - (-2)) = (x + 2)$, $(x - 1)$, and $(x - 3)$ 2. **Multiplicity:** At each of the three x-intercepts, the graph passes directly through.

32 [Intermediate] • Target: Visualizing inverse functions as reflections

The graph of the function $f(x) = -(x - 3)^2 + 4$ is shown in the xy-plane below. What is the maximum value of $f(x)$?

a. -5

b. 3

c. 4 ✓

d. 7

The maximum value of a function refers to the greatest output value (or y-value) that the function attains across its domain. The given quadratic function is written in vertex form, $f(x) = a(x - h)^2 + k$, where $a = -1$, $h = 3$, and $k = 4$. 1. For any real number x, the squared term is non-negative: $(x - 3)^2 \geq 0$. 2. Multiplying by -1 reverses the inequality: $-(x - 3)^2 \leq 0$. 3. Adding 4 to both sides gives: $-(x - 3)^2 + 4 \leq 4$, which means $f(x) \leq 4$ for all x. When $x = 3$, the funct.

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33 [Intermediate] • Target: Domain and range of composite and inverse functions

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the xy-plane with vertex $(2, 9)$ and y-intercept $(0, 5)$. A linear function is defined by $g(x) = mx + 1$. If the graph of g passes through the vertex of the graph of f, what is the other x-coordinate where the graphs of f and g intersect?

a. -4

b. -2 ✓

c. -1

d. 1

We solve the problem in three steps: 1. **Determine the quadratic function $f(x)$:** The vertex is given as $(h, k) = (2, 9)$, so the function has the form: $f(x) = a(x - 2)^2 + 9$ Since the graph passes through the point $(0, 5)$: $5 = a(0 - 2)^2 + 9 \Rightarrow 4a + 9 = 5 \Rightarrow 4a = -4 \Rightarrow a = -1$ Expanding $f(x)$:
$f(x) = -(x - 2)^2 + 9 = -(x^2 - 4x + 4) + 9 = -x^2 + 4x + 5$ 2. **Determine the linear function $g(x)$:** The line $g(x) = mx + 1$ passes through the vertex $(2, 9)$.

34 [Intermediate] • Target: Applying composite and inverse functions in context

The graph of the quadratic function $f(x)$ is shown in the figure, with its vertex at $(1, 8)$ and an x-intercept at $(3, 0)$. A second function g is defined by $g(x) = f(x - 2) + k$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, what is the value of k?

a. -10

b. -6

c. 6

d. **10** ✓

Step 1: Write the equation of $f(x)$ in vertex form. Since the vertex is $(h, k_v) = (1, 8)$, the quadratic function has the form: $f(x) = a(x - 1)^2 + 8$ Step 2: Use the known point $(3, 0)$ to solve for a.
$f(3) = a(3 - 1)^2 + 8 = 0 \Rightarrow 4a + 8 = 0 \Rightarrow a = -2$ Thus, $f(x) = -2(x - 1)^2 + 8$. Step 3: Apply the condition that $g(x)$ passes through $(0, 0)$. $g(0) = f(0 - 2) + k = 0 \Rightarrow f(-2) + k = 0$ Step 4: Evaluate $f(-2)$.
$f(-2) = -2(-2 - 1)^2 + 8 = -2(-3)^2 + 8 = -2(9) + 8 = -10$ Step.

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35 [Intermediate] • Target: Using the horizontal line test

The quadratic function $f(x) = ax^2 + bx + c$ is graphed in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(3, -6)$, and the graph passes through the point $(1, 2)$. What is the value of b ?

a. -12 ✓

b. -6

c. 6

d. 12

1. Write the quadratic function in vertex form: $f(x) = a(x-h)^2 + k$. 2. Given the vertex $(h, k) = (3, -6)$, the function is $f(x) = a(x-3)^2 - 6$. 3. Use the given point $(1, 2)$ on the graph to find a: $2 = a(1-3)^2 - 6$ $2 = 4a - 6 \Rightarrow 4a = 8 \Rightarrow a = 2$. 4. Expand the equation to standard form $ax^2 + bx + c$: $f(x) = 2(x^2 - 6x + 9) - 6 = 2x^2 - 12x + 18 - 6 = 2x^2 - 12x + 12$. 5. Comparing coefficients gives $b = -12$. Common error: A student who ignores the scale factor a might cal.

36 [Intermediate] • Target: Composite functions with restricted domains

Consider the function $f(x) = 4\cos^2(x) + b\sin(x) + c$ where b and c are real constants with $b > 0$. If the maximum value of $f(x)$ is 10 and the minimum value of $f(x)$ is 1, what is the value of c?

a. 5 ✓

b. $\frac{11}{2}$

c. 9

d. 4

To find the unknown parameters, we first express $f(x)$ in terms of a single trigonometric function using the identity $\cos^2(x) = 1 - \sin^2(x)$: $f(x) = 4(1 - \sin^2(x)) + b\sin(x) + c = -4\sin^2(x) + b\sin(x) + c + 4$ Let $u = \sin(x)$. Since $x \in \mathbb{R}$, the variable u is restricted to the interval $[-1, 1]$. We analyze the quadratic function: $g(u) = -4u^2 + bu + c + 4, \quad u \in [-1, 1]$ The graph of $g(u)$ is a downward-opening parabola with its vertex at: $u_v = -\frac{b}{2(-4)} = \frac{b}{8}$ S.

37 [Intermediate] • Target: Composing functions

The function $f(x) = a\cos(bx) + d$, where $a, b,$ and d are positive constants, has a period of π, a maximum value of 7, and satisfies $f(\frac{\pi}{6}) = 4$. What is the value of $f(\frac{\pi}{3})$?

a. -2 ✓

b. 1

c. 4

d. -5

We determine the unknown parameters $a, b,$ and d step by step: 1. **Find b using the period:** The period of $f(x) = a\cos(bx) + d$ is given by $\frac{2\pi}{b}$. Setting this equal to π: $\frac{2\pi}{b} = \pi \Rightarrow b = 2$ 2. **Use the maximum value to relate a and d:** Since $a > 0$, the maximum value occurs when $\cos(bx) = 1$: $a(1) + d = 7 \Rightarrow a + d = 7$ 3. **Use the given point $f(\frac{\pi}{6}) = 4$:** Substitute $b = 2$ and $x = \frac{\pi}{6}$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Decomposing a composite function** • **Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs**

Consider the rational function $f(x) = \frac{x^2 - 2x + c}{x^2 + 2x + c}$ where c is a real constant. If the domain of f is the set of all real numbers and the maximum value of $f(x)$ is 4, what is the value of c?

a. $\frac{25}{9}$ ✓

b. $\frac{9}{4}$

c. 3

d. $\frac{16}{5}$

We are given the rational function $f(x) = \frac{x^2 - 2x + c}{x^2 + 2x + c}$. **Step 1: Domain constraint** For the domain of f to be all real numbers, the denominator $x^2 + 2x + c$ must have no real roots. Therefore, its discriminant must be strictly negative: $\Delta_{denom} = 2^2 - 4(1)(c) = 4 - 4c1$ **Step 2: Range and maximum value via discriminant method** Let $y = f(x)$. Because $x^2 + 2x + c > 0$ for all $x \in \mathbb{R}$, we can multiply through by the denominator.

39 **[Challenging]** • **Target: Verifying inverse functions** • **Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions**

A function f is defined for all real numbers x such that $xeq0$ and $xeq1$. If $f(x) + 2f\left(\frac{1}{1-x}\right) = 3x$ for all permissible values of x, what is the value of $f(3)$?

a. $\frac{20}{9}$ ✓

b. $\frac{7}{3}$

c. $\frac{16}{9}$

d. 3

To find $f(3)$, we use the cyclic substitution property of the transformation $T(x) = \frac{1}{1-x}$. 1. Substitute $x = 3$ into the given functional equation: $f(3) + 2f\left(\frac{1}{1-3}\right) = 3(3) \Rightarrow f(3) + 2f\left(-\frac{1}{2}\right) = 9 \quad - - - \text{ (Equation 1)}$ 2. Next, evaluate the equation at $x = -\frac{1}{2}$: $\frac{1}{1-(-1/2)} = \frac{1}{3/2} = \frac{2}{3}$.

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40 [Challenging] • Target: Visualizing inverse functions as reflections • Cross-Lesson: 4.03 Parametric Equations & Modeling

Which of the following statements describes the end behavior of the polynomial function $f(x) = -5x^4 + 2x^3 - 7x + 3$?

a. As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to \infty$, $f(x) \to -\infty$ ✓

b. As $x \to -\infty$, $f(x) \to \infty$, and as $x \to \infty$, $f(x) \to -\infty$

c. As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to \infty$, $f(x) \to \infty$

d. As $x \to -\infty$, $f(x) \to \infty$, and as $x \to \infty$, $f(x) \to \infty$

To determine the end behavior of a polynomial function, examine its leading term, which dominates the function's values for very large positive and negative values of x. 1. Identify the leading term: The highest power of x is 4, so the leading term is $-5x^4$. 2. Analyze the degree and leading coefficient: - The degree is $n = 4$, which is an even integer. - The leading coefficient is $a = -5$, which is negative. 3. Determine the limits as $x \to \pm\infty$: - As $x \to \infty$.

41 [Challenging] • Target: Domain and range of composite and inverse functions • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A parabola has its vertex at the origin $(0, 0)$ and passes through the point $(2, -4)$. If the directrix of the parabola is a horizontal line, what is the equation of the directrix?

a. y = 4 ✓

b. y = -4

c. x = 4

d. x = -4

The vertex of the parabola is at $(0, 0)$. Since the directrix is a horizontal line, the axis of symmetry is vertical, and the equation of the parabola is of the form $x^2 = 4py$. The parabola passes through the point $(2, -4)$. Substituting these coordinates into the equation, we get $2^2 = 4p(-4)$, which simplifies to $4 = -16p$. Solving for p, we find $p = \frac{4}{-16} = -\frac{1}{4}$. The equation of the parabola is $x^2 = 4(-\frac{1}{4})y$, or $x^2 = -y$. The directrix of a parabola of the form $x^2 = 4py$ i.

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42 [Challenging] • Target: Applying composite and inverse functions in context • Cross-Lesson: 1.02 Rational Functions & Asymptotes

A factory produces two types of components, A and B, using two machines, M1 and M2. The time (in minutes) required to produce one unit of each component on each machine is given in the table below: | Component | Machine M1 | Machine M2 | |---|---|---| | A | 2 | 3 | | B | 4 | 1 | The factory has 480 minutes of M1 time and 240 minutes of M2 time available per day. Let x be the number of units of component A produced per day and y be the number of units of component B produced per day. Suppose the profit from selling one unit of component A is 5 and the profit from selling one unit of component B is 8. The factory wants to maximize its total daily profit. Formulate this problem as a linear programming problem, defining the objective function and the constraints.

a. **Maximize $P = 5x + 8y$ subject to $2x + 4y \le 480$, $3x + y \le 240$, $x \ge 0$, $y \ge 0$.** ✓

b. Maximize $P = 8x + 5y$ subject to $2x + 3y \le 480$, $4x + y \le 240$, $x \ge 0$, $y \ge 0$.

c. Maximize $P = 5x + 8y$ subject to $4x + 2y \le 480$, $y + 3x \le 240$, $x \ge 0$, $y \ge 0$.

d. Maximize $P = 5x + 8y$ subject to $2x + 4y \le 240$, $3x + y \le 480$, $x \ge 0$, $y \ge 0$.

The objective is to maximize the total daily profit. The profit from selling x units of component A is $5x$ and the profit from selling y units of component B is $8y$. Therefore, the objective function is $P = 5x + 8y$. The constraints are based on the available machine time. For Machine M1, producing x units of A takes $2x$ minutes and producing y units of B takes $4y$ minutes. The total time on M1 must be less than or equal to the available 480 minutes: $2x + 4y \le 480$. For Machine M2, producing x un.

43 [Challenging] • Target: Using the horizontal line test • Cross-Lesson: 1.03 Rates of Change & Concavity

The graph of the polynomial function $f(x) = x^3 + ax^2 + bx - 6$ has a local maximum at $x = -3$ and a local minimum at $x = 1$. Find the values of a and b.

a. $a = -3, b = 9$

b. $a = 3, b = -9$ ✓

c. $a = -6, b = -3$

d. $a = 6, b = 3$

For a polynomial function to have local extrema at specific x-values, its derivative must be zero at those points. The derivative of $f(x) = x^3 + ax^2 + bx - 6$ is $f'(x) = 3x^2 + 2ax + b$. Since there are local extrema at $x = -3$ and $x = 1$, we have $f'(-3) = 0$ and $f'(1) = 0$. Substituting $x = -3$ into $f'(x)$: $3(-3)^2 + 2a(-3) + b = 0$ $3(9) - 6a + b = 0$ $27 - 6a + b = 0$ (Equation 1) Substituting $x = 1$ into $f'(x)$: $3(1)^2 + 2a(1) + b = 0$ $3 + 2a + b = 0$ (Equation 2) Now we have a system of two line.

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44 [Challenging] • Target: Composite functions with restricted domains • Cross-Lesson: 2.01 Exponential Functions & Growth Models

The graph of the polynomial function $f(x) = x^3 - ax^2 + bx - 6$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. Determine the value of $a + b$.

a. 5

b. **-5** ✓

c. 1

d. -1

To find the local extrema of a polynomial function, we need to find its critical points by taking the derivative and setting it to zero. The derivative of $f(x)$ is $f'(x) = 3x^2 - 2ax + b$. Since $f(x)$ has local extrema at $x = -1$ and $x = 3$, these are the roots of $f'(x) = 0$. Thus, we can write $f'(x)$ in factored form: $f'(x) = k(x - (-1))(x - 3) = k(x + 1)(x - 3)$ for some constant k. Expanding this, we get $f'(x) = k(x^2 - 2x - 3)$. Comparing the coefficient of x^2 in $f'(x) = 3x^2 - 2ax + b$ and.

45 [Challenging] • Target: Composing functions • Cross-Lesson: 2.02 Logarithmic Functions & Properties

The function f is defined for all real numbers $xeq -2$ by $f(x) = \frac{4x-1}{x+2}$ What is the value of $f^{-1}(3)$?

a. -5

b. 3

c. 5

d. 7 ✓

By definition of the inverse function, if $f^{-1}(3) = x$, then $f(x) = 3$. Set the function equal to 3 and solve for x: $\frac{4x-1}{x+2} = 3$ Multiply both sides by $x + 2$: $4x - 1 = 3(x + 2)$ $4x - 1 = 3x + 6$ Subtract $3x$ from both sides and add 1: $4x - 3x = 6 + 1$ $x = 7$ Thus, $f^{-1}(3) = 7$. Common Misconceptions: - Choice A (-5) arises from a sign error when clearing the denominator as $3(x - 2)$, yielding $4x - 1 = 3x - 6 \Rightarrow x = -5$. - Choice B (3) results from forgetting to dist.

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